Sahaj Bhadja

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Summary

Application software engineer with 2+ years delivering internal and customer-facing systems across the SDLC. Experienced in Python, Java, JavaScript, React, REST APIs, SQL, AWS, Azure, requirements analysis, testing, debugging, documentation, process improvement, and production support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python APIs**, **Azure SQL** services, and React internal applications for 500+ users, translating stakeholder requirements into maintainable workflows while reducing production latency from 1.2 seconds to 350 milliseconds.
- Implemented 5,000+ monthly workflows through explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving observability, recovery, maintainability, operational resilience, and systematic production issue diagnosis.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **React** interfaces and **Flask REST services** handling 10,000+ daily requests, implementing customer workflows, authentication, progress states, and reusable components while resolving production defects and supporting 150+ users.
- Created multithreaded processing for 1,000+ files with validation, **AWS RDS** persistence, progress monitoring, and failure recovery, reducing turnaround 60% while maintaining responsive behavior under concurrent production load.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed scalable **Python services and REST integrations** using object-oriented patterns, filtering, pagination, caching, AWS Glue, S3, and SQL, improving production response times 30% and processing efficiency 68%.
- Maintained 20+ automated unit and integration suites with **Jenkins**, PyTest, and Mockito, accelerating technical troubleshooting, release verification, and reliable cloud application deployments across Windows, Linux, and macOS.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular **React and Node.js** features across eight microservices with authentication, role-based access, responsive interfaces, and real-time updates, processing 2 million+ Azure SQL records through documented integrations.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through **Jenkins and Azure**, reducing release time 50% while improving service reliability, deployment consistency, and operational visibility.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **React Native** clients for iOS and Android with Firebase persistence, real-time state synchronization, image storage, push notifications, and secure sessions across a mobile platform handling 50,000+ records.
- Developed **Flask REST APIs** for bookings, pricing, and payments, delivering sub-200-millisecond responses and reliable external webhook-driven integrations that improved confirmation speed 40% across customer-facing mobile application workflows.

Skills

Languages & Applications: Python, Java, JavaScript, TypeScript, SQL, HTML5, CSS3, React, Node.js, Flask, FastAPI, object-oriented programming, responsive applications
APIs & Data: REST APIs, web services, WebSockets, webhooks, microservices, PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, Firebase, AWS RDS, relational databases, schema design, query optimization
Cloud & Delivery: AWS, Azure, S3, AWS Glue, Docker, Kubernetes, Jenkins, CI/CD, Git, authentication, role-based access, workflow automation, production deployment, monitoring
SDLC & Quality: requirements analysis, technical specifications, Agile, iterative development, unit testing, integration testing, functional testing, debugging, troubleshooting, code review, documentation, process improvement, production support, stakeholder collaboration

Projects

HMIS | Live | *TypeScript, React, REST APIs, Appwrite, Sentry, Twilio*
- Built responsive **healthcare workflows** for appointments, records, and consent-based access using React, REST APIs, authentication, Twilio notifications, and Sentry monitoring, supporting 100+ daily interactions while reducing administrative workload 30%.

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases with schema-aware query generation, structured validation, optimized execution, metadata exploration, and Docker-backed deployment across modular **Java and TypeScript services**.

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a responsive platform supporting 50,000+ users and one million transactions, integrating **secure APIs**, real-time WebSocket synchronization, MongoDB persistence, reusable interfaces, Kubernetes deployment, and automated release workflows.